

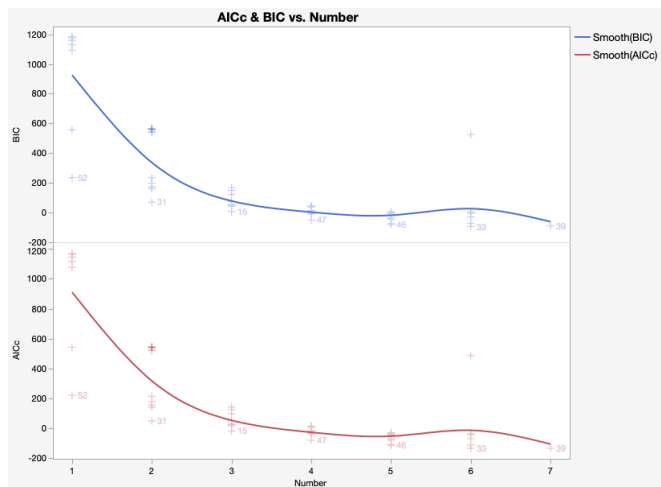
Factors Influencing the Median Home Value Across U.S. Metropolitan Areas

ILRST 2130 Project Component 5

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In this final component, model selection techniques are applied to identify the most appropriate multiple regression model for explaining variation in log-transformed median home value across U.S. metropolitan areas. Using the same set of predictor variables examined in previous components, an all-possible regression approach was conducted in JMP to evaluate models of varying sizes. The model performance was assessed using different selection criteria, including AIC, BIC, adjusted R^2 , residuals mean squared error (RMSE), and Mallows' C_p . The goal of this analysis is to determine which subset of predictors provides the best balance between explanatory power and model simplicity, ultimately selecting a final model that is both statistically sound and interpretable.

AIC & BIC vs Number of Predictors



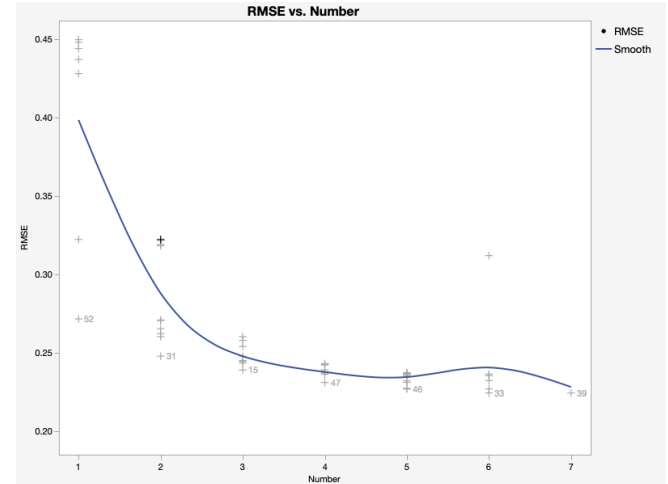
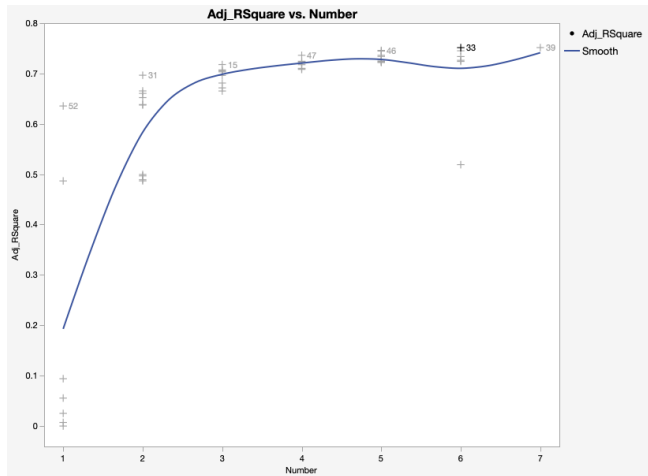
The AIC and BIC values both decrease sharply as the number of predictors increases from one to approximately four or five variables. After about five predictors, the reduction in AIC and BIC becomes much smaller and begins to level off, suggesting diminishing improvement in model performance as additional variables are added. Based on the AIC and BIC criteria, **Model 33** provides the best fit because it has the lowest values for both statistics (AIC = -133.06). This model

includes six predictors, indicating that the six-variable model performs best according to these criteria. However, the improvement between the five- and six-predictor models is relatively small compared to the earlier decreases like **Model 46** (AIC = -113.54, BIC = -79.74). This suggests that most of the improvement in model fit occurs before five predictors are included.

Adjusted R^2 vs Number of Predictors

The adjusted R^2 values increase as predictors are added to the model, but the improvement becomes very small for larger models. The highest adjusted R^2 occurs for **Model 39** (Adj R^2 = 0.7514), which includes seven predictors. However, this value is nearly identical to **Model 33** (Adj R^2 = 0.7513), which contains only six predictors. **Model 46**, which includes five

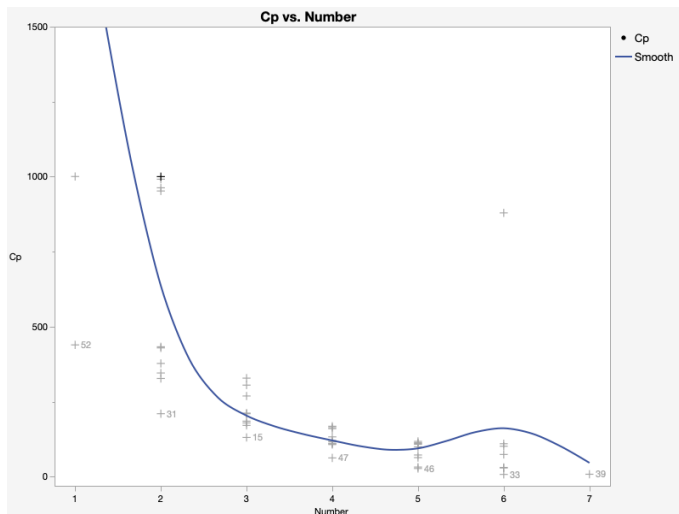
predictors, also has a similar adjusted R^2 value (0.7458). This indicates that adding additional predictors beyond five or six variables provides little improvement in explanatory power after accounting for model complexity. Therefore, the five- or six-predictor models appear to provide a good balance between model fit and parsimony.



RMSE vs. Number of Predictors

The smallest RMSE value also occurs for both **Model 33** and **Model 39** (RMSE = 0.2243), indicating that this model provides the most accurate predictions among the candidate models. Since this model includes six predictors, it suggests that adding predictors up to this point improves prediction accuracy. However, something to note is that Model 46 (RMSE = 0.2268) with only 5 predictors also gives a very similar level of predictive accuracy compared to Model 33.

Mallows' C_p vs. Number of Predictors



The lowest C_p value occurs for **Model 33** ($C_p = 7.4081$), which again corresponds to the six-predictor model. This suggests that Model 33 provides the best balance between model fit and complexity according to the C_p criterion. Other nearby models have slightly larger C_p values but still perform similarly. For example, **Model 46** has a higher C_p value of 27.06, but it includes only five predictors, making it a simpler model.

Because the improvement in C_p from adding an additional predictor is relatively small, Model 46 remains a reasonable alternative due to its greater simplicity and interpretability.

Final Candidate Model(s): Model 33

Model 33 includes **six** predictors:

- *median household income*
- *median age*
- *unemployment rate*
- *percent renters*
- *percent bachelors plus*
- *vacancy rate*

According to the model selection criteria, Model 33 performs very well. It has the lowest RMSE (0.2243), as well as the lowest AIC (-133.06), BIC (-94.46), and C_p value (7.41) among the candidate models. These statistics suggest that Model 33 provides the best overall model fit and predictive accuracy. Because it minimizes several selection criteria, this model can be considered the strongest model in terms of statistical performance.

Final Candidate Model(s): Model 46

Model 46 includes **five** predictors:

- *median household income*
- *median age*
- *percent renters*
- *percent bachelors plus*
- *vacancy rate*

Although Model 46 has slightly higher values for the selection criteria (RMSE = 0.2268, AIC = -113.54, BIC = -79.74, C_p = 27.06), its performance is very similar to Model 33 while using one fewer predictor. The only variable excluded from Model 33 is *unemployment rate*. Because the improvement in model fit from including this additional predictor is relatively small, Model 46 was selected as the preferred model. This model provides nearly the same predictive performance while remaining simpler and easier to interpret. Based on the model selection criteria and simplicity, Model 46 was selected as the final regression model.

Conclusion

Overall, the model selection analysis suggests that models with approximately five to six predictors provide the best fit according to the AIC, BIC, RMSE, and Mallows' C_p criteria. Although Model 33 achieved the lowest values for several statistics, the improvement compared to Model 46 was relatively small. Following the principle of parsimony, Model 46 was selected as the final model because it achieves similar predictive performance while using fewer predictors. The final model includes *median household income*, *median age*, *percent renters*, *percent bachelors plus*, and *vacancy rate*, which together explain a substantial portion of the

variation in log median home value. These results suggest that socioeconomic and housing composition variables play an important role in explaining variation in median home values across metropolitan areas.